

Rania Atmani Garmavich

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Fall 2026 Computer Engineering Co-op | Embedded Systems, Firmware, Hardware Test, Validation

EDUCATION

University of British Columbia Okanagan | BAsc Computer Engineering, Co-op Program | Kelowna, BC | Expected Apr 2028

Average: 86% | Strongest CS area: data structures and algorithm analysis | Relevant coursework: Data Structures, OOP, Digital Logic Design, Signals and Communication Systems, Electric Circuits, Numerical Methods, Computer Engineering Design Studio

Douglas College | Engineering Foundations Certificate | Honour Roll | CGPA: 3.26/4.33 | Sept 2024 - Apr 2025

TECHNICAL SKILLS

Languages: Java (strongest), C++ (intermediate), Python (working proficiency)

Computer science: data structures and algorithm analysis; strongest in theoretical reasoning, tracing behavior, complexity, and edge-case analysis

Embedded/hardware: ESP32, Raspberry Pi, GPIO, HC-SR04 ultrasonic sensor, SG90 servo, VEX robotics, sensor interfacing, relay-control logic, serial/debug output

Testing/tools: validation runs, data logging, wiring/setup notes, acceptance criteria, Git/GitHub, PlatformIO, PowerShell, technical documentation

TECHNICAL PROJECTS

ESP32 HVAC Thermostat Prototype | Independent project | C++/PlatformIO, ESP32, sensors, relay-control logic

- Modeled an HVAC control problem by building an ESP32 prototype that reads temperature data, documents relay behavior, and records setup assumptions for a realistic field-service use case.
- Improved reproducibility by implementing dependency configuration, serial/debug checks, secrets-template files, wiring notes, and setup steps, enabling safer clone/review workflows.
- Demonstrated firmware-test readiness by connecting microcontroller I/O, sensor/control logic, validation planning, and troubleshooting notes in one working embedded project.

Ultrasonic Radar Prototype | CMPE 246 group course project | Raspberry Pi, HC-SR04, SG90 servo, browser visualization

- Supported a 0-180 degree sensing demo by documenting GPIO mapping, sensor/servo setup, verification strategy, screenshots, and data-log evidence for final report/demo review.
- Improved traceability by connecting requirements, acceptance criteria, risk notes, and reproducibility material, helping explain how 3 hardware components produced mapped readings.
- Strengthened hardware-test communication by turning noisy prototype behavior into clear evidence, limits, and next-step recommendations for evaluators.

Autonomous Mobile Robot | Douglas College capstone team project | C++/VEXcode, embedded control, sensors

- Programmed control routines for a robot required to navigate a structured course, execute turns, avoid obstacles, and stop at checkpoints under repeated course-run tests.
- Improved reliability by debugging sensor behavior and physical constraints, connecting code changes to hardware outcomes and strengthening validation habits across code and devices.
- Contributed to integration decisions by comparing expected route behavior against observed performance, which improved troubleshooting during team testing.

Onsite Heating Pro | Independent project | React/TypeScript, service records, documentation

- Translated real HVAC workflows into a field-service app covering 6 record types: customer history, job notes, invoices, payments, repeat visits, and follow-up tracking.
- Added embedded-team relevance by documenting practical constraints and user-facing technical notes, strengthening communication between field behavior, records, and project structure.

EMPLOYMENT HISTORY

Onsite Heating & Cooling Ltd. | Assistant Technician | New Westminster, BC | Sept 2024 - Jan 2025

- Accomplished faster service-call readiness by preparing tools, checking parts, and documenting job status, which gave technicians clearer setup information before/after visits.
- Improved traceability by recording technician notes, follow-up items, and safety-aware observations, which supported repeat-visit context and cleaner maintenance records.
- Converted HVAC workflow issues into engineering requirements by mapping field constraints to thermostat/web-tool features, which improved sensor, control, and documentation decisions.

All West Heating Ltd. | Admin and Field Assistant | Coquitlam, BC | Oct 2022 - Apr 2023

- Completed 600+ hours supporting installations, maintenance scheduling, invoice records, parts lists, safety checks, and customer follow-up.
- Improved field coordination by organizing parts information, appointment details, and service records, which kept repeat visits traceable and easier to plan.
- Built equipment-centered troubleshooting awareness by supporting service documentation, which strengthened attention to safety, constraints, and evidence.

LEADERSHIP AND INVOLVEMENT

UBCO Motorsports | Student Engineering Design Team Member

- Strengthen hardware/software judgment by participating in design discussions on subsystem interfaces, wiring constraints, safety/reliability tradeoffs, testing expectations, and integration planning.
- Support team communication by tracking design-review notes and asking clarifying questions, connecting component constraints to documentation before build/test work.